

Instructor: Date:

Learner: Regional Branch:

Program: Center:

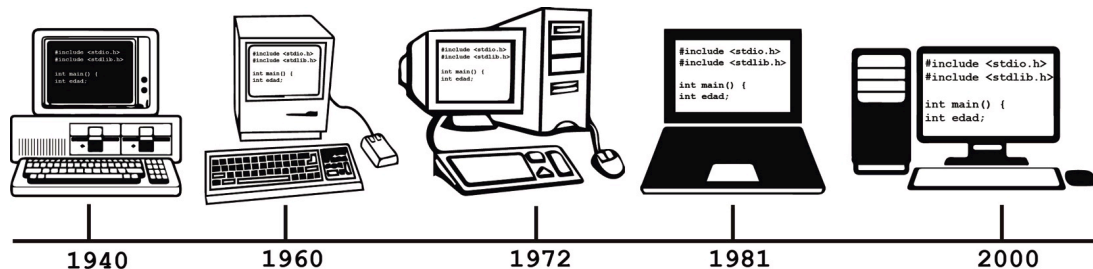
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Let's Start!

Going Back in Time to Analyze Software.



Learning Outcome:

Identify information about the history of software and its characteristics in simple spoken and written situations.

Ask your partner:

- What do you know about the first computers?
- How do you imagine the first software?

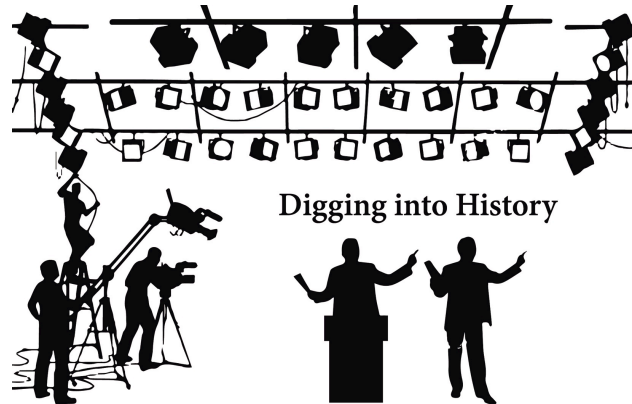
Can Do Statements:

- I can talk about essential characters in the origins of software.
- I can mention ideas about devices that preceded the modern computers.
- I can exchange information about the origins of software and hardware with my colleagues.



Let's Explore!

A Talk Show About History.



Instructions: Read the text below.

- Jack:** Welcome to another episode of Digging into History. Today, our guest is going to talk about software before computers. Thank you for coming, Mrs. Ross.
- Mr. Ross:** Thanks for inviting me. Well, I want to start by talking about essential characters in the origins of software. Most people know that the first computer was the ENIAC, made in the USA at the end of World War II. However, what people don't know is that the concept of software was already old by then. It started in the 19th century in England.
- Jack:** But how? The idea of a computer was too crazy at that time, wasn't it?
- Mr. Ross:** Not really. One important character was Charles Babbage. He was one of the greatest thinkers of the time, and he was a mathematician with many ideas. He was too privileged since he was able to invest a lot of time working with what could be, in theory, the first computer.
- Jack:** I know that, at that time, the most common devices were mechanical calculators, they were useful for mathematical problems, and I guess they were the closest thing to a computer.
- Mr. Ross:** But they weren't! Babbage worked on something called the Difference Engine. This device was too complex for some people, and the idea was to develop a theory of equations. Unfortunately, it was too expensive for Babbage to complete his work, and he had some problems working with his engineer. He also worked on something famously called the Analytical Engine, which included the use of punched cards.
- Jack:** What can you tell us about the origins of software and hardware?
- Mr. Ross:** Dear colleagues, it is important to know that those concepts are older than computers and that many people in history have contributed to the technology of our times.



Let's Practice!

Let's go back in time, how was software back then?

Instructions: Circle the correct option to complete the sentences.

1. Back in the 19th century, mechanical calculators (was / were) common, while programmed computers were not yet fully invented.
2. The Analytical Engine (was / were) very innovative because it was based on mathematical computation.
3. The Difference Engine (was / were) a complex machine.



Let's Practice!

They were, and still are, important characters in the world of computer science.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. Charles Babbage was...
2. Augusta Ada Byron was the...
3. Charles Babbage was an...

COLUMN B

- a. ... a genius of mathematics and computers.
- b. ... important character in the development of the programmable computer.
- c. ... first programmer. Even her name, Ada, is a programming language.



Extension Activity

Let's get more information on software history.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information about the early days of software.
- Design a time line in which you include the most important events in software history.
- Prepare a presentation to describe your time line.
- Share you work with your classmates and instructor.

Self-Assessment

Yes No

- I look for information about the early days of software. ☐ ☐
- I design a time line with all the information I found. ☐ ☐
- I share my time line with my classmates and instructor. ☐ ☐



Target Words

- **character** (noun): a relevant person in a story
e.g. One important character was Charles Babbage.
- **punched card** (noun): system that involved making holes on a card to get digital information
e.g. He also worked in something famously called the Analytical Engine which included the use of punched cards.
- **engineer** (noun): a professional who works with numbers and equations in different fields of mechanics
e.g. Unfortunately, it was too expensive for Babbage to complete his work and he had some problems working with his engineer.
- **expensive** (adjective): something that is worth a lot of money
e.g. Unfortunately, it was too expensive for Babbage to complete his work.
- **famously** (adverb): being recognized because you did something remarkable
e.g. He also worked in something famously called the Analytical Engine which included the use of punched cards.
- **mathematician** (noun): someone who works with mathematics
e.g. He was one of the greatest thinkers of the time, and he was a mathematician with many ideas.



Learning Points

Simple Past Tense of Verb Be, Was in Affirmative Statements.

You use the **verb to be** in past to talk about actions that already happened.

The **verb to be** expresses the current state of people, objects or situations. When conjugated in the past, it describes events that are no longer happening. Events that started and finished in the past.

The past of the **verb to be** changes depending on the pronoun. Take a look at the following chart:

Pronoun	Verb to be
I	was
You	were
He / She / It	was
We	were
They	were

To write affirmative statements with the **verb to be in past**, we can use the following grammatical structure:

Subject + **was/were** + complement

Let's see a couple of examples:

*The first computer **was** difficult to operate.*

*Charles Babbage **was** a very intelligent mathematician.*

Examples

- *Most people know that the first computer was the ENIAC.*
- *The concept of software was already old by then.*
- *One important character was Charles Babbage.*
- *He was one of the great thinkers of the time, and he was a mathematician with many ideas.*





Learning Points

Verb Be with Too and Adjectives.

You use the **verb to be** to talk about present or past statements. When you use **too + adjective**, you add an extra description to the statement.

In the previous learning point, we observed how the verb to be is conjugated in the past and how we use was and were to talk about the past. Now, we are able to use the adverb too plus an adjective to say that the characteristic is excessive.

We can follow the next grammatical structure:

Subject + verb to be (in past or present) + **too + adjective** + complement

Let's see an example:

The first computer was **too difficult** to operate.

Notice how the adverb **too** provides an extra meaning to the sentence. In this case, we could say that not many people had the ability to operate the first computer because it was very difficult to do it.

We can use **too** with any kind of **adjective**, let's see:

*Understanding software is **too hard**.*

Examples

- *The idea of a computer was too crazy at that time, wasn't it?*
- *He was too privileged since he was able to invest a lot of time working with what could be, in theory, the first computer.*
- *This device was too complex for some people.*

